



Thermal analysis of a cylindrical lithium-ion battery

Xiongwen Zhang*

Experimental Power Grid Centre, Institute of Chemical & Engineering Sciences, A*STAR, 1 Fusionopolis Way, Singapore 138632

ARTICLE INFO

Article history:

Received 19 August 2010

Received in revised form 11 October 2010

Accepted 16 October 2010

Available online 23 October 2010

Keywords:

Lithium-ion battery

Thermal analysis

Ohmic heat

Active polarization heat

Reaction heat

ABSTRACT

This work investigates the heat generation characteristics of a cylindrical lithium-ion battery. The battery consists of the graphite, LiPF_6 of the propylene carbonate/ethylene carbonate/dimethyl carbonate (PC/EC/DMC) solution, and spinal as anode, electrolyte and cathode, respectively. The coupled electrochemical–thermal model is developed with full consideration of electrolyte transport properties as functions of temperature and Li ion concentration. A truly conservative finite volume numerical method is employed for the spatial discretization of the model equations. Three types of heat generation sources including the ohmic heat, the active polarization heat and the reaction heat are quantitatively analyzed for the battery discharge process. The ohmic heat is found to be the largest contribution with around 54% in the total heat generation. About 30% of the total heat generation in average is ascribed to the electrochemical reaction. The active polarization contributes the least comparing to the ohmic heat and reactions heat. The results also show that the Li ion concentration and its gradient in electrolyte are the main factors giving the effect on the heat generations of active polarization and electrolyte electric resistance. The raised temperature in the battery discharge is positive related with the thickness of both separator and electrodes.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

The development of microgrid and electric vehicles (EV) has been accompanied by advances in battery technology as to energy storage. Particularly high expectations are being placed on advances in high-performance lithium-ion batteries (such as high energy density, high open circuit voltage, and low self-discharge). Although lithium-ion batteries have been extensively applied in the portable electronic devices, a few key issues have to be investigated for the practical applications of EV and microgrid with large capacity of energy storage. Heat control and management is one of the most important issues in the lithium-ion batteries as high temperature will lower charge/discharge efficiency and lower the battery life, or even cause safety problem.

The thermal characteristics of lithium-ion batteries have been intensively studied through both of the model analysis and of the experimental measurement. Specially, the coupled thermal–electrochemical models have been developed by the researchers [1–3] based on the electrochemical model of Doyle et al. [4] with an addition of energy balance equation. These coupled models are efficient tools to be used for analyzing and understanding the thermal behaviors in the lithium-ion battery design and optimization as well as the thermal management for the appli-

cations. In the battery operation, the heat generation is mainly ascribed to the electrochemical reactions, the active polarization and the electric conductive resistance. The behaviors of these thermal generation factors were analyzed by Sato [5] through the thermodynamic tool. The experimental data of the electrochemical reaction entropy change as reported by [6,7] indicates that the reaction heat in lithium-ion battery varies substantially with state of charge (SOC). The SOC was found to be one of the most important parameters with the effect on the heat generation [8]. The thermodynamic response of lithium-ion battery depends on the heat generation, heat dissipation and heat capacity. The operating temperature is determined by the balance between the heat generation and the heat dissipation. Thus, the thermal model analysis is required to properly deal with the boundary conditions to avoid the distorted results. Harchard et al. [9] presented a cylindrical lithium-ion battery thermal model. This model considered the cylinder as a serial of concentric rings. A 1-dimensional numerical scheme was established at the radial direction. The dissipating heat transfer coefficient is made up of all heat flow in term of conduction, convection and radiation. However, this thermal model does not consider the effect of variant Nusselt number (forced convection case) and battery geometry. In order to properly calculate the heat dissipation on the boundary, Chen et al. [10,11] introduced a few semiempirical equations with both the Nusselt number and the battery geometry effect integrated in the dissipating heat transfer coefficient. Temperature gradient in lithium-ion battery is another important subject for the cell design and optimization. As an effec-

* Tel.: +65 6408 2912; fax: +65 6466 7389.

E-mail address: zhang.xiongwen@ices.a-star.edu.sg

Nomenclature

a	specific surface area (m^{-1})
a_k	parameter of parabolic function Eq. (25) for control volume k
A_i	fit parameter for voltage curve of Eq. (13) (V)
A_{sf}	area of side skin surface for cylindrical battery (m^2)
A_{ef}	area of end cover for cylindrical battery (m^2)
b_k	parameter of parabolic function Eq. (25) for control volume k
c	Li ion concentration in the electrolyte (mol m^{-3})
c_s	Li solid-phase concentration in the electrodes (mol m^{-3})
c_t	maximum Li solid-phase concentration in the electrodes (mol m^{-3})
C_p	specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
b_k	parameter of parabolic function for control volume k
d_k	parameter of parabolic function Eq. (25) for control volume k
D	diffusion coefficient of Li in electrolyte ($\text{m}^2 \text{s}^{-1}$)
D_s	diffusion coefficient of Li in solid phase ($\text{m}^2 \text{s}^{-1}$)
f	boundary condition parameter of thermal model ($\text{W m}^{-2} \text{K}^{-n-1}$)
f_1	boundary condition parameter of thermal model ($\text{W m}^{-2} \text{K}^{-n-1}$)
F	Faraday's constant ($=96487 \text{ C mol}^{-1}$)
g	function of Eq. (1d)
h	heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
H	battery characteristic length (m)
H_r	height of roll (m)
i	current density in electronic conductive electrodes (A m^{-2})
i_0	exchange current density (A m^{-2})
i_e	current density in the electrolyte (A m^{-2})
I	cell charge/discharge current (A)
j_n	electrochemical reaction rate ($\text{mol m}^{-2} \text{s}^{-1}$)
k	parameter of electrochemical reaction rate Eq. (17) ($\text{A mol}^{3/2} \text{m}^{5/2}$)
l	length of components (m)
L	length of spiral ring (m)
m	boundary condition parameter of thermal model
M	parameter of Eqs. (27a) and (27c)
N	number of revolutions of the spiral
P	parameter of Eqs. (27a) and (27b)
$\dot{q}$	heat generation rate (W m^{-2})
Q	parameter of Eqs. (27a) and (27d)
r	cylindrical radius of battery (m)
r_2	parameter of Eqs. (1a) and (1c) (m)
R	ideal gas constant ($=8.314 \text{ J mol}^{-1} \text{K}^{-1}$)
R_s	solid particle radius of electrodes (m)
S	source term of concentration balance equation, Eq. (24) ($\text{mol m}^{-2} \text{s}^{-1}$)
Δs	electrochemical reaction entropy change
t	time (s)
T	temperature (K)
T_0	initial temperature (K)
T_∞	ambient temperature (K)
U	output voltage (V)
V	volume (m^3)
w	thickness of spiral roll (m)
x	coordinate along the thickness direction of cell components (m)
z	dimensionless Li concentration ($z = c_s/c_t$)

Greek letters

α	transfer coefficient of the electrochemical reactions
δ	thickness of cell components (m)
ε	volume fraction of porosity
ε_f	volume fraction of filler
ε	emissivity
η	voltage loss (V)
θ	time relax factor as Eq. (27a)
κ	ion conductivity (S m^{-1})
λ	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
ν	electrolyte salt thermodynamic factor
ξ_i	fit parameters for voltage curve
χ_i	fit parameters for voltage curve
ρ	density (kg m^{-3})
σ	electronic conductivity (S m^{-1})
τ	tortuosity
ζ	correction coefficient
ϕ	potential in the solid phase (V)
ϕ_e	potential in electrolyte (V)
ϕ_{oc}	open-circuit potential (V)
ϕ_0	fit parameter for voltage curve of Eq. (13) (V)
Θ	Stefan–Boltzmann constant ($=5.6704 \times 10^{-8} \text{ W m}^{-2} \text{K}^{-4}$)

Subscript

a	anode
acc	anode current collector
act	active polarization
av	average
b	bulk
c	cathode
ccc	cathode current collector
e	interface ($k + (1/2)$) of control volume k
i	index of spiral roll layer
j	index of components layer
k	index of control volume
liq	liquid electrolyte
ohm	ohmic loss
rea	reactions
s	separator
sol	solid electrodes
su	surface
w	interface ($k - (1/2)$) of control volume k

Superscript

k	the k th cell
n	time index

tive, efficient and economic tool, numerical simulation has been widely used for analyzing the temperature distribution from 1-D to 3-D by the researchers [12–14].

Most of current thermal models or experimental measurements focused on the overall heat generation of battery discharge. As we know, the overall heat generation of battery discharge is consisted of electrochemical reaction heat, ohmic heat and active polarization heat. These heats are produced within the negative electrode, electrolyte and positive electrode during the discharge of battery. The above three types of heat generation sources are spatial variation in the cell. The discharge electrochemical reaction includes the lithium extraction and intercalation, which take place at the anode/electrolyte interface and cathode/electrolyte interface, respectively. Therefore, the electrochemical reaction heat is divided into lithium extraction heat and lithium intercalation

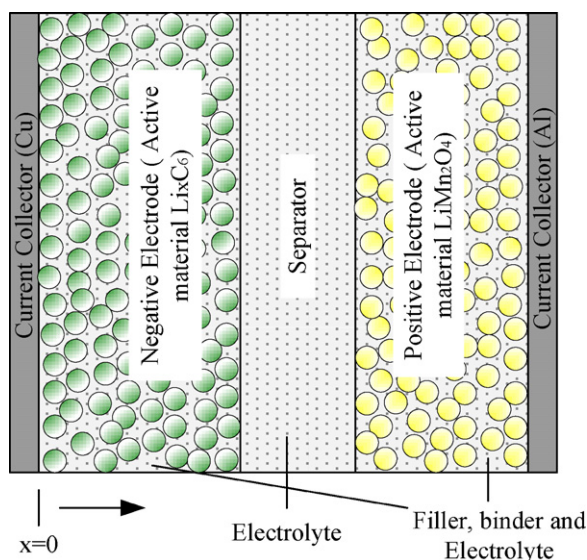


Fig. 1. Schematic of a lithium-ion cell.

heat. Similarly, the active polarization heat is also composed of the terms of lithium extraction and intercalation. The ohmic heat is contributed by the electron-conduction resistance (anode and cathode) and ion-conduction resistance (electrolyte). Understanding the thermal behavior of these heat generation sources would offer much valuable information to develop the battery thermal management strategies as well as cell design, optimization and operation.

The emphasis of present work is to analyze different heat generation sources in the discharge of a cylindrical lithium-ion battery. The cell consists of lithium manganese oxide ($\text{Li}_y\text{Mn}_2\text{O}_4$) positive electrode and graphite mesocarbon microbead (MCMB) 2528 negative electrode. LiPF_6 in a solvent mixture of propylene carbonate/ethylene carbonate/dimethyl carbonate (PC/EC/DMC) is used as the electrolyte. A thermal model is developed based on 1-D electrochemical equations coupled with a cell-lumped energy equation. The temperature and concentration dependent parameters of electrolyte transport properties including diffusion coefficient, conductivity, thermodynamic factor, etc., were referred to the experimental characterization given by Valøen and Reimers [15]. The temperature dependence is important because generally, the rate capability of a lithium ion cell is improved due to the self-heating during the discharge/charge proceeds. The Li ion concentration dependence is important because the gradients in the salt concentration are formed during the discharge/charge operation. The concentration gradients give much effect on the conductivity of electrolyte and active polarization of electrochemical reactions. A truly conservative control volume method based on piecewise second-degree polynomial approximation scheme is established for the spatial discretization of model equations.

2. Model development

2.1. Principle and schematic of Li-ion battery

Fig. 1 gives the schematic of a lithium-ion cell. It consists of negative electrode (anode), separator and positive electrode (cathode). Both the electrodes and the separator are porous structure. The electrode is composed of active material, conductive filler, polymer binder and electrolyte. The active materials of anode and cathode are graphite MCMB 2528 and $\text{Li}_y\text{Mn}_2\text{O}_4$, respectively. The electrolyte is LiPF_6 in a mixture of PC (10 vol.%), EC (27 vol.%), and DMC (63 vol.%) [15]. The porous separator is composed of liquid elec-

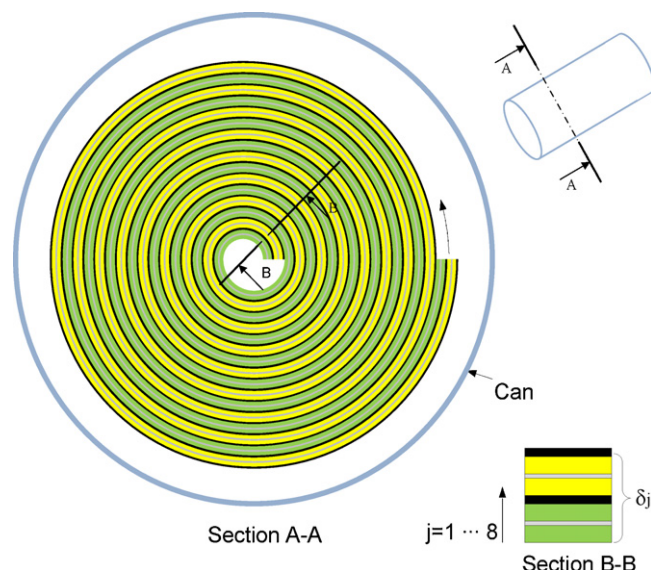
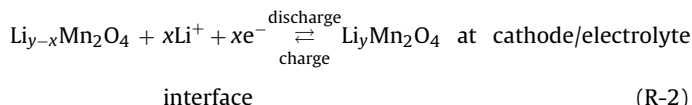
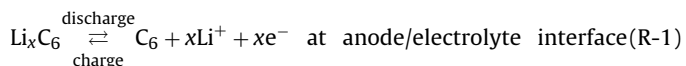


Fig. 2. Cross section of a cylindrical lithium-ion battery. A spiral roll section B–B of is composed of components including anode (δ_1, δ_3), current collector of anode (δ_2), separator (δ_4, δ_8), cathode (δ_5, δ_7), and current collector of cathode (δ_6).

trolyte and p(VdF-HVP) [16,17]. The electrochemical reactions are expressed as



A typical cylindrical lithium battery is spiral rolls cell as shown in Fig. 2. Thin layers of cathode, separator, current collector, and anode are rolled up on central mandrel and inserted into a cylindrical can. The gaps are filled with liquid electrolyte. The section B–B in Fig. 2 gives the detail configuration of the rolling layer as described in [10]. The basic design remains unchanged on substitution of one electrode material for another, although the layer thickness might change for different designs. The length of spiral ring can be calculated by

$$L_{i,j} = \frac{\pi}{w} \left\{ r_2 g(r_2) - r_{i,j} g(r_{i,j}) + \left(\frac{w}{2\pi} \right)^2 [\ln(r_2 + g(r_2)) - \ln(r_{i,j} + g(r_{i,j}))] \right\} \quad (1a)$$

$$w = \sum_{j=1}^8 \delta_j \quad (1b)$$

$$r_2 = r_{i,j} + w \quad (1c)$$

$$g(r) = \sqrt{r^2 + \left(\frac{w}{2\pi} \right)^2} \quad (1d)$$

where i is the index of spiral rolling layers at the radial direction, j is the index of components within the rolling layers range from 1 to 8 as shown the section B–B in Fig. 2. If we consider each 360° spiral ring as a cell and the revolution number is N , then a cylindrical battery is equivalent to $2N - 1$ cells with connection each other in parallel. The component circumferences for the k th cell are given by

$$l_a^k = L_{i,3}, \quad l_s^k = L_{i,5}, \quad l_c^k = L_{i,5}, \quad l_{acc}^k = L_{i,2}, \quad l_{ccc}^k = L_{i,6},$$

$$\text{where } k \text{ is odd number and } i = \frac{k+1}{2} \quad (2a)$$

$$l_a^k = L_{i+1,1}, \quad l_s^k = L_{i,8}, \quad l_c^k = L_{i,7}, \quad l_{acc}^k = L_{i+1,2}, \quad l_{ccc}^k = L_{i,6},$$

$$\text{where } k \text{ is even number and } i = \frac{k}{2} \quad (2b)$$

If a cylindrical battery is considered as a lumped single cell, the lengths of cell components are calculated by

$$\begin{aligned} l_a &= L_{13} + \sum_{i=2}^N (L_{i,1} + L_{i,3}), \quad l_s = L_{N,4} + \sum_{i=1}^{N-1} (L_{i,4} + L_{i,8}), \quad l_c \\ &= L_{N,5} + \sum_{i=1}^{N-1} (L_{i,5} + L_{i,7}), \quad l_{acc} = \sum_{i=1}^N L_{i,2}, \quad l_{ccc} = \sum_{i=1}^N L_{i,6} \end{aligned} \quad (3)$$

3. Mathematics model

The basic battery physics occurs in a wide range of length and time scales. It includes the electrochemical kinetics, solid-phase lithium transport, lithium ion transport in electrolyte, charge conservation/transport, energy conservation/dissipation and heat transfer. The model used to describe the ionic transport in the electrolyte is based on multicomponent diffusion, mass transfer, and concentrated solution theory [18]. The lithium transport balance equation in the electrolyte was given by [4]

$$\varepsilon \frac{\partial c}{\partial t} = \nabla \cdot \left\{ \left[\varepsilon D \nabla c + \frac{i_e (1 - t_+^0)}{F} \right] \right\} \quad (4)$$

$$j_n = \frac{\nabla \cdot i_e}{aF} \quad (5)$$

where a is specific surface area of porous defined as $3(1 - \varepsilon - \varepsilon_f)/R_s$ for spherical particles. The potential difference along the current flow in the electrolyte is a function of the lithium ion concentration and current density as follows [19]:

$$\nabla \phi_e = \frac{2RT}{F} \nabla [\nu \ln c] - \frac{i_e}{\kappa} \quad (6)$$

The transport properties including the salt diffusion coefficient D , the conductivity κ , and the salt thermodynamic factor ν are the temperature and concentration dependence. A fully characterized electrolyte used in present work was reported by Valøen and Reimers [15] as given blow

$$D_b = 1 \times 10^{-4} 10^{-4.43(54.0/(T-229.0-0.005c)) - 2.2 \times 10^{-4} c} \quad (7)$$

$$\kappa_b = 1 \times 10^{-4} c (-10.5 + 0.074T - 6.69 \times 10^{-5} T^2 + 6.68 \times 10^{-4} c - 1.78 \times 10^{-5} cT + 2.8 \times 10^{-8} cT^2 + 4.94 \times 10^{-7} c^2 - 8.86 \times 10^{-10} c^2 T)^2 \quad (8)$$

$$\begin{aligned} \nu &= 0.601 - 0.24 \sqrt{10^{-3} c} \\ &+ 0.982 [1 - 0.0052(T - 294.0) \sqrt{10^{-9} c^3}] \end{aligned} \quad (9)$$

The mass transport of electrolyte is reduced due to the irregular pore and channel in the porous structure of electrodes and separator. A geometric parameter so-called tortuosity τ is introduced in the model. The salt effective diffusion coefficient and conductivity of electrolyte are calculated by $D_b \varepsilon / \tau$ and $\kappa_b \varepsilon / \tau$, respectively. The solid-phase lithium transport in the particles of electrodes is described with parabolic approximation at radial direction [20,21]

$$\frac{\partial c_{s,av}}{\partial t} = -3aj_n \quad (10)$$

$$c_{s,av} - c_{s,su} = \frac{aj_n R_s}{5D_s} \quad (11)$$

The solid-phase diffusion coefficient of Li in cathode is considered to be constant in present work. Temperature dependence for solid-phase diffusion coefficient of Li in anode is estimated by an Arrhenius correlation as follows [3]:

$$D_{s,a} = 1.4523 \times 10^{-13} \exp \left[\frac{68025.7}{R} \left(\frac{1}{318} - \frac{1}{T} \right) \right] \quad (12)$$

The open-circuit potential of the electrode materials at reference temperature T_0 is fit to the function

$$\phi_{oc}(x) = \phi_0 + \sum_{i=0}^n \frac{A_i}{1 + \exp \xi_i(z - \chi_i)} \quad (13)$$

where z is c_s/c_t . The parameter ϕ_0 is 0.005769 and 0.6185 for the anode Li_xC_6 and cathode $\text{Li}_y\text{Mn}_2\text{O}_4$, respectively. Other parameters of Eq. (13) used in present paper are given in Table 1 [15]. The potential in solid-phase field is given by

$$\nabla \phi = -\frac{i}{\sigma} \quad (14)$$

The total ohmic voltage loss η_{ohm} is the addition of the electrical resistances within the anode, cathode and electrolyte. It is calculated by Eqs. (6) and (14). The energy barrier, which is called active polarization, exists between the electronic and the ionic conductor of the electrochemical reaction. The active polarization occurs at both electrodes in the whole ranges of charge/discharge current. It is generally expressed by the Butler–Volmer equation:

$$i = i_0 \left\{ \exp \left(\alpha_a \frac{F \eta_{act}}{RT} \right) - \exp \left[-\alpha_c \frac{F \eta_{act}}{RT} \right] \right\} \quad (15)$$

where $\alpha_a + \alpha_c = 1$

In this paper, α_a and α_c are set to be 0.5. Then

$$\eta_{act} = \frac{2RT}{F} \sinh^{-1} \left(\frac{i}{i_0} \right) \quad (16)$$

The exchange current densities i_0 for the both electrodes have the form

$$i_0 = k c^{\alpha_a} (c_t - c_{s,su})^{\alpha_a} c_{s,su}^{\alpha_c} \quad (17)$$

The temperature dependant of open-circuit potential is approximated by first order of Taylor's expansion. Thus, the output voltage has the expression as

$$U = [\phi_{oc,c} + (T - T_0) \Delta s_{rea,c}] - [\phi_{oc,a} + (T - T_0) \Delta s_{rea,a}] - \eta_{ohm} - \eta_{act} \quad (18)$$

where the entropy changes $\Delta s_{rea,a}, \Delta s_{rea,c}$ for Li_xC_6 at anode with

$$\Delta s_{rea,a} = \Delta s_{rea,c} \quad (8)$$

reaction Eq. (R-1) and $\text{Li}_y\text{Mn}_2\text{O}_4$ at cathode with reaction Eq. (R-2) are determined by linearly interpolation based on the measured samples given by the Refs. [6] and [22], respectively. The heat flows in axial and angular directions are assumed to be neglected. Hence, a 1-D heat flow inside the cylindrical battery is dominated by conduction in radial directions, which is given by

$$\rho C_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(\lambda r \frac{\partial T}{\partial r} \right) + q \quad (19)$$

where the heat generation rate for a cell is defined by

$$q = \frac{1}{V} (q_{rea} + q_{act} + q_{ohm}) \quad (20)$$

$$q_{rea} = (T - T_0) (\Delta s_{rea,a} - \Delta s_{rea,c}) \quad (21)$$

$$q_{act} = \eta_{act} I \quad (22)$$

$$q_{ohm} = \eta_{ohm} I \quad (23)$$

Table 1Open-circuit potential fit parameters for the anode Li_xC_6 and cathode $\text{Li}_y\text{Mn}_2\text{O}_4$ [15].

i	$A_{i,a}$	$\xi_{i,a}$	$\chi_{i,a}$	$A_{i,c}$	$\xi_{i,c}$	$\chi_{i,c}$
1	53.03	116.4	-0.015616	111.1	526.9	-0.003787
2	0.3863	57.80	0.05939	0.8014	27.84	-0.07037
3	0.7979	31.91	0.2025	0.2002	7.841	0.4807
4	0.03996	21.76	0.5146	0.1505	14.60	0.9493
5	0.04007	19.17	0.8803	0.3580	213.1	0.9965
6	0.1534	39.54	1.008	0.4480	426.3	1.005
7	-0.1045	36.25	1.026	0.1941	1133	1.010
8	—	—	—	1.176	28.96	1.211
9	—	—	—	1.013	56.22	1.297

3.1. Numerical approach

In present model, the equation of lithium ion transport in electrolyte (Eq. (4)) is solved with 1-D from anode to cathode within a cell using numerical iteration method. A truly conservative finite volume method for 1-D reaction-diffusion problems was reported by Ramos [23]. For 1-D model, the combined Eqs. (4) and (5) can be written as follows:

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) + S \quad (24)$$

where $S = aj_n(1 - t_+^0)/\varepsilon$. The variable c is assumed to be a parabola distribution as

$$c(x) = a_k(x - x_k)^2 + b_k(x - x_k) + d_k \quad (25)$$

where $x[x_{k-1}, x_{k+1}]$. By solving the equations $c(x_{k-1}) = c_{k-1}$, $c(x_k) = c_k$ and $c(x_{k+1}) = c_{k+1}$, the coefficients a_k , b_k , d_k are obtained by

$$a_k = \frac{1}{2\Delta x}(c_{k-1} - 2c_k + c_{k+1}) \quad (26a)$$

$$b_k = \frac{1}{2\Delta x}(c_{k+1} - c_{k-1}) \quad (26b)$$

$$d_k = c_k \quad (26c)$$

Then, the Eq. (24) can be discretized by integration method in $[x_w, x_e]$ with the expression as [23]:

$$\frac{1}{\Delta t}(P^{n+1} - P^n) = \theta M^{n+1} + (1 - \theta)M^n + \theta Q^{n+1} + (1 - \theta)Q^n \quad (27a)$$

$$P = \frac{1}{24}(c_{k+1} + 22c_k + c_{k-1}) \quad (27b)$$

$$M = \frac{D_e}{\Delta x^2}(c_{k+1} - c_k) - \frac{D_w}{\Delta x^2}(c_k - c_{k-1}) \quad (27c)$$

$$Q = \frac{1}{24}(S_{k+1} + 22S_k + S_{k-1}) \quad (27d)$$

Neumann's boundary conditions are applied in present work for Eq. (24). The discrete formulations of the boundary conditions for Li concentration were given by Ref. [23]. The energy conservation equation as thermal model is applied to a lumped cell. For a cylindrical battery, if it is regarded as $2N - 1$ spiral ring cells connected in parallel, then Eq. (19) is solved by numerical iteration with $2N$ nodes in the radial direction. A finite difference method as given in Ref. [9] can be used for the discretization of energy conservation equation. The thermal boundary condition includes both convection and radiation heat removal as given by [11]. As shown in Table 2, the heat dissipation from the end covers of the cylindrical battery is included in the lumped thermal model. A correction coefficient ζ is proposed to estimate the thermal resistance between the end covers and spiral roll. In this work, the ζ is set to be 0.5. The boundary equations are specified as shown in Table 2.

Table 2

Boundary conditions.

Variables	Boundary conditions
c	$\frac{\partial c}{\partial x} = 0$ at $x = 0$ and $x = \delta_a + \delta_s + \delta_c$
ϕ	$i = \frac{I}{l_{acc}H_r}$ at $x = 0$, $i = -\frac{I}{l_{ccc}H_r}$ at $x = \delta_a + \delta_s + \delta_c$
ϕ_e	$i_e = 0$ at $x = 0$ and $x = \delta_a + \delta_s + \delta_c$
T	For 1-D thermal model: $\frac{\partial T}{\partial r} = 0$ at $r = 0$, $-\lambda \frac{\partial T}{\partial r} = h(T - T_\infty)$ at $r = R$ For lumped thermal model: $\rho C_p \frac{\partial T}{\partial t} = q + \frac{h(A_{sf} + \zeta A_{ef})}{V}(T - T_\infty)$ where $h = h_c + h_r = f T - T_\infty ^m + \epsilon\theta(T^2 + T_\infty^2) + (T - T_\infty)$, $f = \frac{f_1}{H^m}$; $f_1 = 1.485088$, $m = 0.25$ for $H > 0.152$ m and $f_1 = 0.941145$, $m = 0.35$ for $H < 0.152$ m.

4. Discharge simulation

The model validation is taken by the existed experimental data. ValØen and Reimers [15] measured the skin temperature of a 65 mm high and 26 mm diameter cylindrical lithium-ion battery. This battery consists of graphite anode, spinal cathode and 0.96 M LiPF_6 concentration in PC/EC/DMC as electrolyte. In present work, we keep the same of the battery sizes and cell components materials as that of Ref. [15]. The C rate is equivalent to the discharge current 3 A for present battery unit. The battery parameters specified in the simulation are given in Table 3. The discharge simulations are implemented with the help of MATLAB software platform.

The validation case is performed on the discharge rate of 5 A (1.7C). The starting temperature is 25 °C. The skin temperature at the end of discharge is obtained to be 39.43 °C by the lumped thermal model simulation with the cutoff voltage 2.5 V. As shown in Fig. 3, the battery can skin temperature given by 1-D thermal model

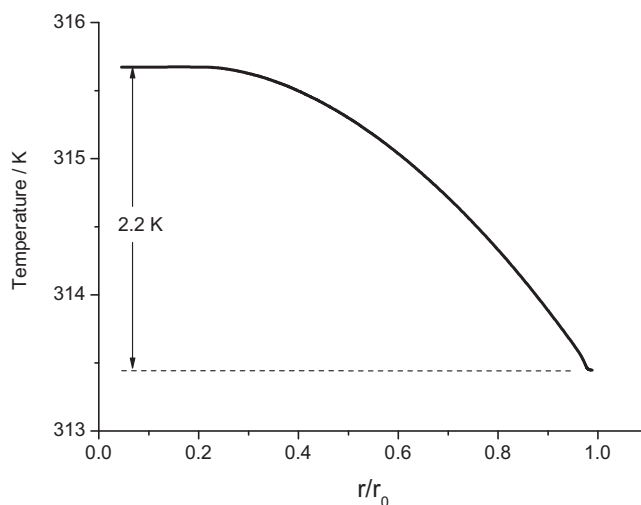


Fig. 3. Temperature distribution along the radial direction at the end of discharge with 1.7C (cutoff voltage 2.5V).

Table 3
Physical parameters of a cylindrical lithium-ion battery.

Parameters	Value			Source
Cylindrical battery geometry parameters				
Diameter (mm)	26			
Height (mm)	65			
Number of roll revolutions	28	d		
Gap between the roll and can (μm)	80	d		
Can thickness (mm)	0.3	a		
Height of roll (mm)	57.2	d		
Cell components parameters				
	Anode	Separator	Cathode	
Thickness (μm)	55	26	86.5	d
Current collector thickness (μm)	15		20	[15]
Porosity	0.262	0.37	0.267	[15]
Tortuosity	2.9	2.8	2.8	[15]
Volume fraction of filler	0.044		0.151	[24]
Solid-phase diffusion coefficient (m ² s ⁻¹)	Eq. (12) [3]	–	1.0e–13[17]	–
Electronic conductivity (S m ⁻¹)	1.0e+02	–	3.8	[17]
Radius of electrode particles (μm)	12.5	–	8.5	[17]
Maximum concentration of active material (mol m ⁻³)	30540	--	22860	[17]
Material properties				
Can density (kg m ⁻³)	7.917e+03			[9]
Electrolyte density (kg m ⁻³)	1.24e+03			e
Li _x C ₆ density (kg m ⁻³)	2.20e+03			[17]
Li _y Mn ₂ O ₄ density (kg m ⁻³)	4.14e+03			[17]
Filler density (kg m ⁻³)	2.20e+03			[17]
Density of filled silica in separator (kg m ⁻³)	2.63e+03			e
Average specific heat of spiral roll (J kg ⁻¹ K ⁻¹)	800			[15]
Can specific heat (J kg ⁻¹ K ⁻¹)	460			[9]
Can thermal conductivity (W m ⁻¹ K ⁻¹)	14			[9]
Average thermal conductivity of spiral roll (W m ⁻¹ K ⁻¹)	0.99			e
Reaction rate constant (A m ^{5/2} mol ^{3/2})	$k_a = 5.7456 \times 10^{-11}$ $k_c = 3.5771 \times 10^{-10}$			e
Surface emissivity	0.8			[9]

a, assumption values; d, design adjustable parameters; e, estimation values.

is 40.4°C . Both of the simulation values are very close to the measurement value 40°C as reported in [15]. Fig. 3 also shows the temperature distribution along the radial direction within the battery. About 2.2°C is found for the temperature difference between the centre and the can skin. This value gives little effect on the accuracy of the simulations for present lithium ion battery. The following discussions are presented based on the lumped thermal model simulation results.

The others two cases are simulated with the discharge rates 0.5C and 1C, respectively. Fig. 4 shows the simulated voltage curves at different discharge rates. The output voltage is opposite to the discharge rate in the whole range. This can be explained by two main reasons. As we know, the reactions at the anode and cathode during the discharge are Li deintercalation and intercalation, respectively.

The Li concentrations in the active material particle present descending distribution in the anode and ascending distribution in the cathode along the radial direction from the particle centre to the reaction interface. The Li concentration gradient increases with increasing the discharge rate. Therefore, the difference of solid-phase Li concentration between the reaction interfaces of anode and cathode is decreased at the higher discharge rate, which gives the smaller open-circuit voltage. Additionally, a higher discharge rate produces larger gradient of Li ion concentration in the electrolyte. According to Eq. (6), the larger Li ion concentration gradient leads to higher voltage loss in the electrolyte.

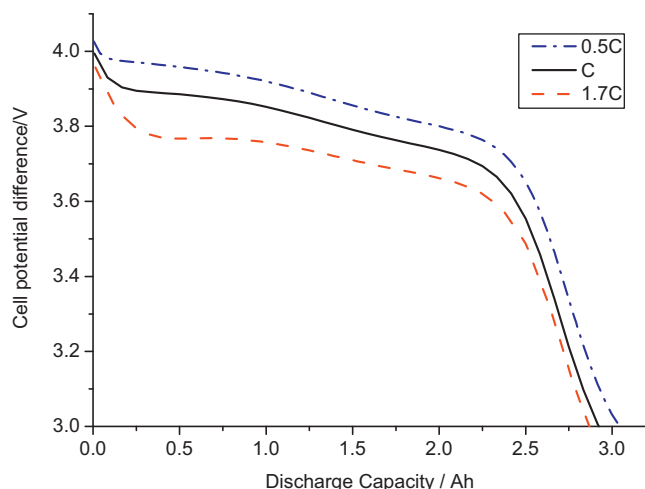


Fig. 4. Cell discharge potential curves at different rates.

5. Results and discussion

The total heat generation during the charge/discharge of lithium ion battery is constituted by three types so-called the reaction heat, the active polarization heat and the ohmic heat. The reaction heat is due to the changes of potential energy in the electrochemical reactions (R-1) and (R-2). Generally, the reaction heat is calculated by the entropy change in the electrochemical reactions. The active polarization refers to the energy barrier that must be overcome for the reaction to proceed. It is accompanied by heat generation occurring at both electrodes. In the following discussion, the reaction heat and active polarization heat are separately computed at anode and cathode. The ohmic Joule heat is composed of the Li ion conductive resistance in the electrolyte and electronic resistive heat in the electrodes.

Fig. 5 shows the battery skin temperature changes at different discharge rates. Obviously, the temperature increases with increasing the discharge rate. The temperature curves for each of discharge rates present rising characteristic overall in the whole range. The

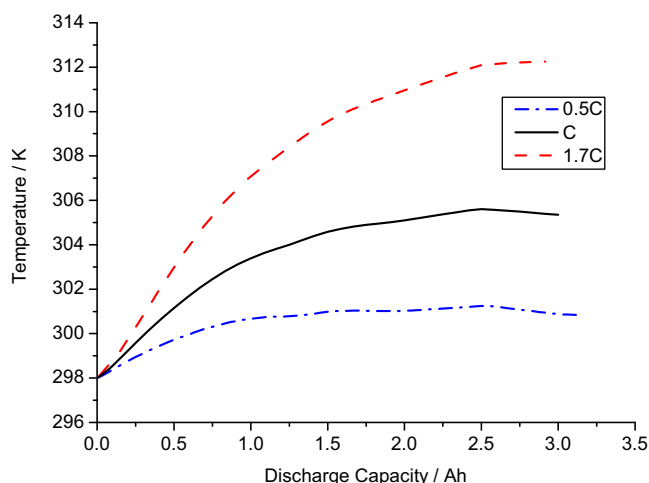


Fig. 5. The skin temperature of the cylindrical lithium-ion battery at different discharge rates.

curve slope goes opposite to the discharged capacity. During the beginning time of discharge, the temperature rises much faster than that of the latter of discharge. Furthermore, as we can see from the Fig. 5, the smaller is the discharge rate, the earlier the battery reaches the thermal balance between the heat generation and heat removal. The magnitudes of the temperature change at the end of discharge for 0.5C, 1C and 1.7C discharge rates are 2.84 K, 7.36 K and 14.43 K, respectively. These data indicate that the raised temperature is almost linear to the discharge rate at the natural convective cooling condition.

The heat sources in the discharge of lithium-ion battery mainly include the electrochemical reaction heat, the active polarization heat and the ohmic heat. Fig. 6 gives the heat generation rates of these three items at the discharge rate of 1C. As shown in Fig. 6, the largest heat source in the discharge comes from the ohmic resistance. The reactions heat is the next to the largest heat source. Comparatively, the active polarization heat is pretty low with about 16% of total heat generation, which is about 0.3 of the ohmic heat or half of the reactions heat in average. If we calculate the average value, the ohmic heat contributes about 54% in the total generated heat. The electrochemical reactions heat is around 30%. The ohmic heat increases significantly after starting the discharge. This is also the main reason that the temperature rises very fast during the beginning time of discharge. In Fig. 6, the reaction heat generation rate presents wave feature in the whole time of discharge.

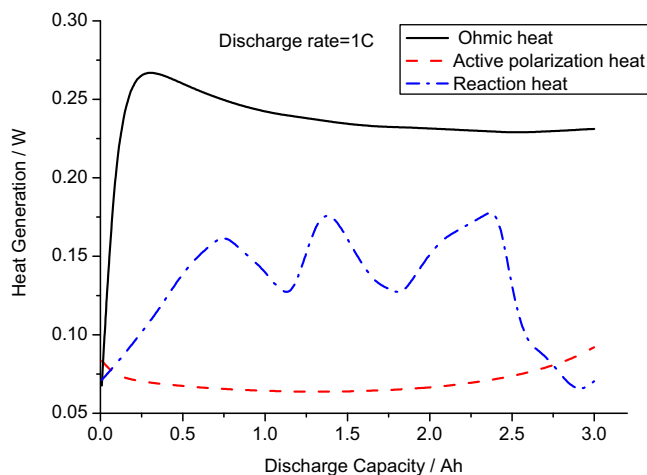


Fig. 6. The constitution of the total heat generation during the discharge of 1C.

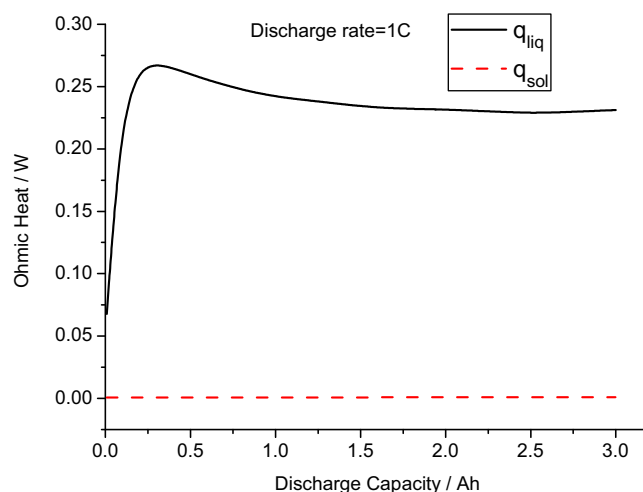


Fig. 7. Ohmic heat generation due to electric conductive resistances in the electrolyte and electrodes during the discharge of 1C.

This is reflected to the small fluctuation of temperature as shown in Fig. 5.

Fig. 7 shows the constitutions of the total ohmic heat at 1C discharge rate for the lithium ion battery. The term q_{liq} denotes the Li ion resistive heat generation in the liquid electrolyte within the anode, separator and cathode. The q_{sol} due to the electronic conductive resistance occurs in the solid electric material of electrodes. The scale of the electronic ohmic heat is as low as 10^{-4} W. It is considered negligible in comparison with the Li ion ohmic heat in the electrolyte. In the beginning time of discharge, the gradient of $LiPF_6$ concentration increases rapidly from zero along the current flow in the electrolyte. This leads to significant increasing of voltage loss (see Eqs. (6) and (9)). The Li ion ohmic heat reaches a peak value in a short time after starting the discharge. Since the electrolyte conductive coefficient increases with the rising of temperature, the q_{liq} value decreases after the peak point with the trend that declines fast at first then gets slow. This trend is agreement with the temperature changes characteristics (see Fig. 5).

The reaction heat varies strongly with the state of charge (SOC) in the electrodes containing the active material. The reaction heat can be either exothermic or endothermic, depending on the entropy of reaction. Fig. 8 shows the changes of reaction heat generation during the discharge of 1C at the anode and cathode, respectively.

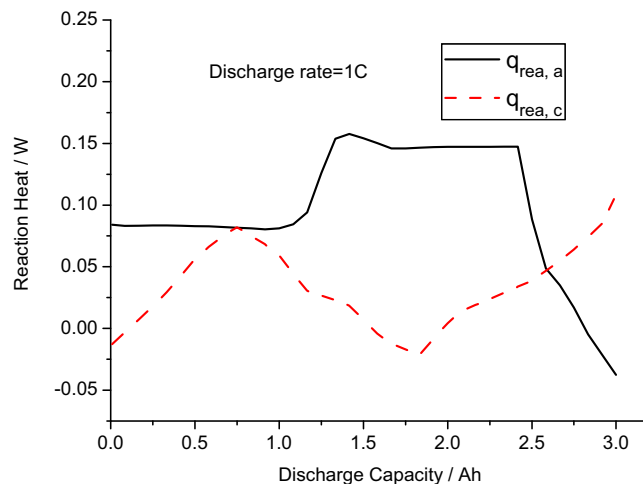


Fig. 8. The electrochemical reactions heat generation at the anode and cathode during the discharge of 1C.

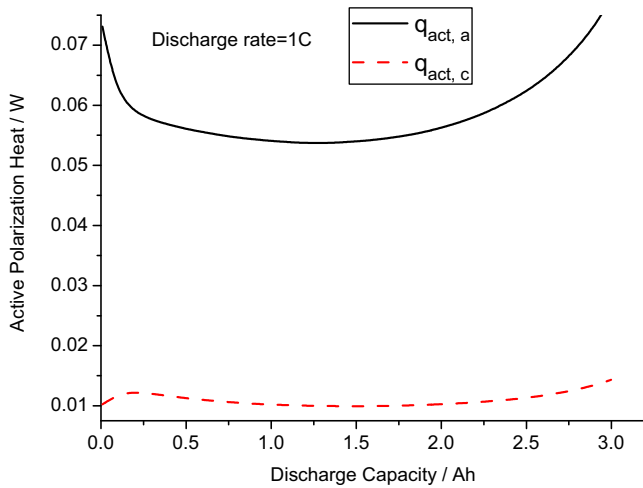


Fig. 9. Active polarization heat generation at the anode and cathode during the discharge of 1C.

The reaction heats generated at the interfaces of anode/electrolyte (Li deintercalation) and cathode/electrolyte (Li intercalation) are computed according to the entropy changes of the backward reactions of (R-1) and (R-2). Both of the curves as shown in Fig. 8 are not regular pattern. Especially, the endothermic effect indicated by the negative values of q_{rea} occurs in the short periods for both of the electrochemical reactions. In the most of time during the discharge, the released heat from the Li deintercalation reaction is more than that of Li intercalation reaction.

Fig. 9 gives the active polarization heat (q_{act}) with the discharge rate 1C. The values of q_{act} for the Li deintercalation reaction at anode are much higher than that of the Li intercalation reaction at cathode. As shown in Fig. 9, the active polarization heat at anode decreases quickly in the beginning discharge period. However, the $q_{act,c}$ goes opposite with $q_{act,a}$ in this period. This can be explained by the effect of Li ion concentration. Fig. 10 shows the distribution of LiPF₆ concentration in the electrolyte across the anode, separator and cathode at the end of discharge. As we know, the initial concentration of LiPF₆ is 0.96 M. During the battery discharge process, the deintercalation reaction at anode releases the Li ion to the electrolyte. The Li ion concentration in the electrolyte of anode increases due to the mass transfer resistance (see Fig. 10). The exchange current density for the deintercalation reac-

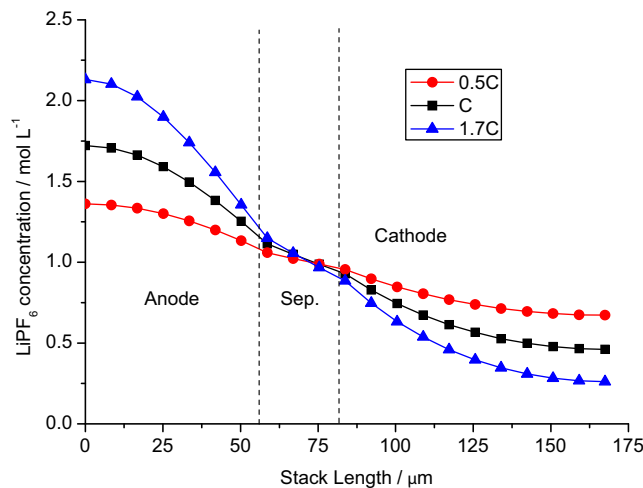


Fig. 10. LiPF₆ concentration profile across the anode/separator/cathode after 3.0 Ah had been discharged from the cell.

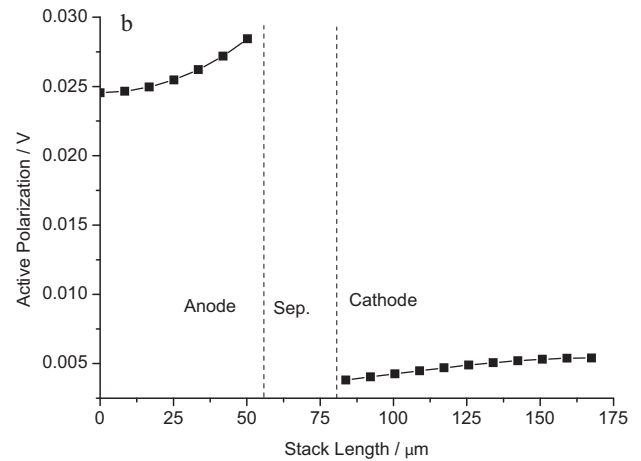
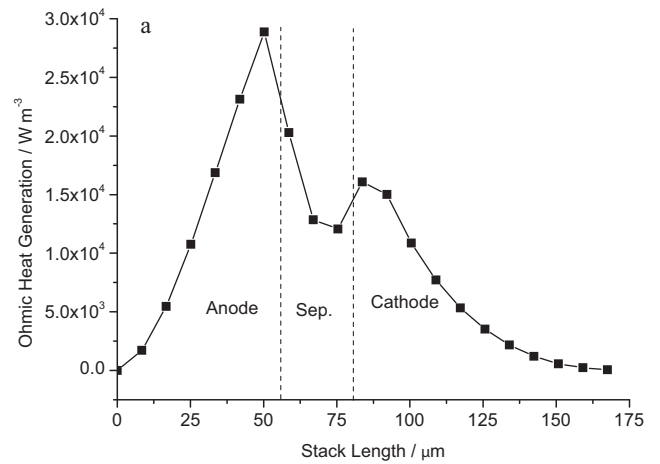


Fig. 11. Spatial distributions of ohmic heat generation and active polarization across the anode/separator/cathode at the end of 3.0 Ah discharged with 1C (a) ohmic heat generation rate (b) active polarization.

tion at anode increases with increasing the Li ion concentration in electrolyte. The higher exchange current density leads to smaller active polarization. With the same reason as shown in Fig. 10, the Li ion concentration in the electrolyte of cathode decreases due to the intercalation reaction during the discharge. The exchange current density for the intercalation reaction at cathode is reduced by the decreased Li ion concentration in electrolyte. The active polarization heat at cathode increases with decreasing the reaction exchange current density. As shown in Fig. 9, both the active polarization heat at anode and cathode grow up before the end of discharge. This is because the Li concentration in the solid particles of active material is close to the maximum concentration (cathode intercalation reaction) or zero (anode deintercalation reaction), which leads to decline the exchange current density (see Eq. (17)).

The Li ion concentration and its gradient in the electrolyte have much effect on both of ohmic heat and of active polarization heat. As shown in Fig. 10, the gradient of Li ion concentration is positive related with the discharge rate. Fig. 11 shows the spatial distribution of ohmic heat generation and active polarization after 3 Ah stored energy is released at 1C discharge rate. The distribution pattern of ohmic heat generation can be explained by the spatial distribution characteristics of Li ion concentration gradient given in Fig. 10. The gradient of Li ion concentration is the most influential factor to the ohmic heat generation in the electrolyte. The ohmic heat generation increases with increasing the gradient of Li ion concentration. As we mentioned in the previous section, the Li

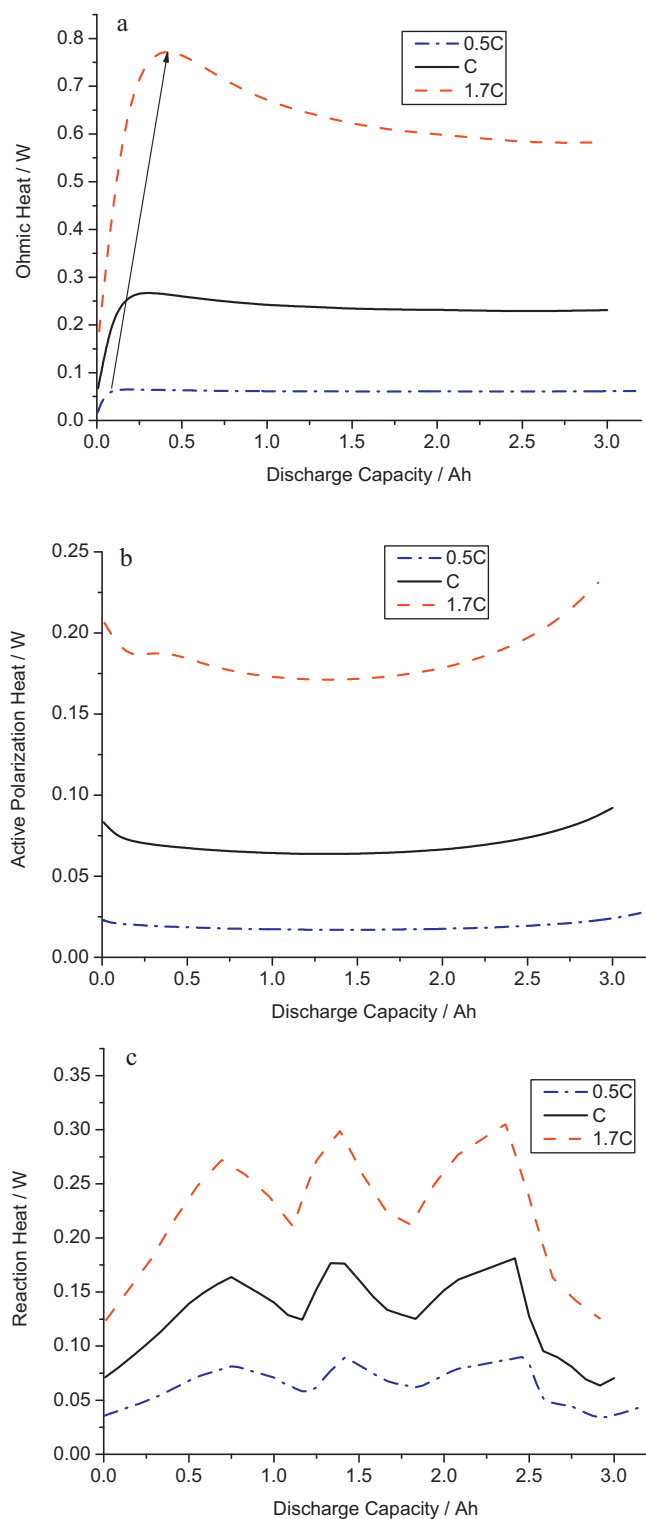


Fig. 12. The heat generation characteristics at different discharge rates (a) ohmic heat generation (b) active polarization heat (c) electrochemical reactions heat.

ion concentration has much effect on the exchange current density, which is one of the most important factors impacting on the active polarization. Combining the Figs. 10 and 11(b), we can find that the active polarizations are negative related with the Li ion concentration.

Fig. 12 shows the heat generation characteristics with various discharge rates. All types of the heat sources including the ohmic heat, the active polarization heat and the electrochemical reaction

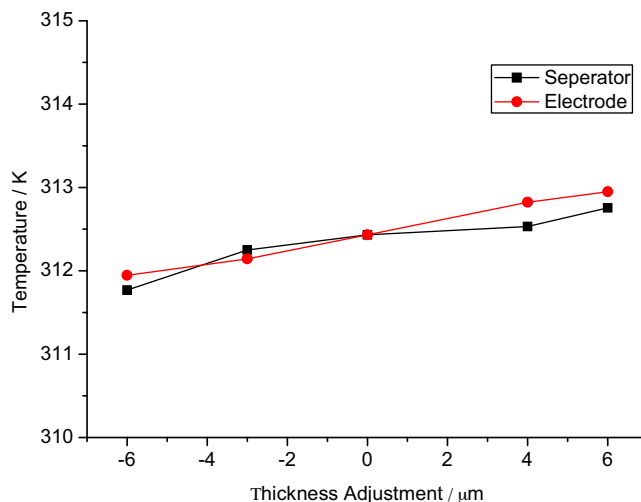


Fig. 13. thickness effect of separator and electrodes on the skin temperature at the end of discharge with 1.7C (cutoff voltage = 2.5 V, original thickness $\delta_a = 55 \mu\text{m}$, $\delta_s = 26 \mu\text{m}$ and $\delta_c = 86.5 \mu\text{m}$).

heat increase with increasing the discharge rate. The characteristics for each type of the heat sources are similar at different discharge rates. As shown in Fig. 12(a), the peak of ohmic heat generation appears later when the discharge rate is increased from 0.5C to 1.7C. Although the side reactions (Li deintercalation/intercalation) could present heat absorption with negative values (see Fig. 8), the overall reaction heat as shown in Fig. 12(c) is always positive values at different rates in the whole ranges. The combined (R-1) and (R-2) is exothermic reaction through the discharge process.

As the ohmic heat produced in electrolyte is the maximum contribution in the total heat generation, here we only present the sensitive analysis of the thickness of separator and electrodes. As given in Table 2, the original thicknesses of anode, separator and cathode are set to be 55 μm , 26 μm and 86.5 μm , respectively. The anode and cathode thickness are kept constant with the original values for the cases of changing separator thickness. Similarly, the separator thickness is set to be 26 μm when the electrode thicknesses are adjusted in the simulations. In order to keep the initial lithium mass ratio between anode and cathode constant as the original cell for the cases of electrodes thickness sensitive analysis, the adjustment magnitudes of electrodes are aligned with the original thickness proportion of anode and cathode. Fig. 13 shows the thickness effect of separator and electrodes on the temperature. For increasing separator or electrodes thickness, both lead to increase the temperature with similar magnitudes during the discharge. As we know from the Figs. 6 and 7, the ohmic heat due to the lithium ion conductive resistance in the electrolyte gives significant effect in the total heat generation during the discharge. The route length of current flow in the electrolyte is increased with increasing the thickness of separator or electrodes, which leads to more ohmic heat generation in the electrolyte.

6. Conclusions

A coupled electrochemical–thermal model was presented for analyzing the thermal characteristics of a cylindrical lithium-ion battery. The electrodes active materials studied in this work are Li_xC_6 and $\text{Li}_y\text{Mn}_2\text{O}_4$, respectively. The electrolyte is composed of 0.96 M LiPF_6 in mixture of PC/EC/DMC. The temperature and Li ion concentration dependence for the transport properties of electrolyte were included in the developed model. A truly conservative finite volume method was used for solving the electrochemical reaction-diffusion model equations. A combined heat transfer coef-

ficient with considering the convective and radiative heat transfer as well as the effect of battery geometries was employed in the thermal boundary condition. The skin temperature of the cylindrical battery obtained by model simulation at the end of the discharge (1.7C) was agreement with the measurement data.

The discharge process was specially simulated using the developed model at the rates of 0.5C, 1C and 1.7C. Three types of heat generation sources including the ohmic heat, the active polarization heat and the reactions heat were separately computed in the whole discharge process. The simulation results show that the ohmic heat is the largest heat generation source, which contributes as much as about 54% of the total heat generation in average. Most of ohmic heat is generated by the reason of Li ion conductive resistance in the electrolyte. The heat generation from the electric resistance of electrodes can be ignored in comparison to the Li ion ohmic heat. The electrochemical reaction heat holds the next largest heat source with average 30% of the total heat generation. The third main heat source comes from the active polarization taking about 16% of the total heat generation. For the discharge process, the active polarization of Li intercalation reaction at anode is much larger than that of Li deintercalation reaction at cathode. The Li ion concentration and its gradient are recognized to be the most important factors with effect on the heat generations of Li ion ohmic resistance and active polarization. The temperature at the end of discharge increases with increasing the thickness for both of separator and electrodes.

References

- [1] C.R. Pals, J. Newman, *J. Electrochem. Soc.* 142 (1995) 3274.
- [2] W.B. Gu, C.Y. Wang, *J. Electrochem. Soc.* 147 (2000) 2910.
- [3] K. Kumaresan, G. Sikha, R.E. White, *J. Electrochem. Soc.* 155 (2008) A164.
- [4] M. Doyle, T.F. Fuller, J. Newman, *J. Electrochem. Soc.* 140 (1993) 1526.
- [5] N. Sato, *J. Power Sources* 99 (2001) 70.
- [6] K.E. Thomas, J. Newman, *J. Power Sources* 119 (2003) 844.
- [7] Q. Huang, M. Yan, Z. Jiang, *J. Power Sources* 156 (2006) 541.
- [8] E.P. Roth, D.H. Dougherty, *J. Power Sources* 128 (2004) 308.
- [9] T.D. Hatchard, D.D. MacNeil, A. Basu, J.R. Dahn, *J. Electrochem. Soc.* 148 (2001) A755.
- [10] S.C. Chen, C.C. Wan, Y.Y. Wang, *J. Power Sources* 140 (2005) 111.
- [11] S.-C. Chen, Y.-Y. Wang, C.-C. Wan, *J. Electrochem. Soc.* 153 (2006) A637.
- [12] L. Song, J.W. Evans, *J. Electrochem. Soc.* 147 (2000) 2086.
- [13] Y. Inui, Y. Kobayashi, Y. Watanabe, Y. Watase, Y. Kitamura, *Energy Convers. Manage.* 48 (2007) 2103.
- [14] K. Onda, T. Ohshima, M. Nakayama, K. Fukuda, T. Araki, *J. Power Sources* 158 (2006) 535.
- [15] L.O. ValØen, J.N. Reimers, *J. Electrochem. Soc.* 152 (2005) A882.
- [16] P. Arora, M. Doyle, R.E. White, *J. Electrochem. Soc.* 146 (1999) 3543.
- [17] P. Arora, M. Doyle, A.S. Gozdz, R.E. White, J. Newman, *J. Power Sources* 88 (2000) 219.
- [18] J.S. Newman, *Electrochemical System*, 2nd edition, Prentice Hall, Englewood Cliffs, NJ, 1991.
- [19] P. Georen, G. Lindbergh, *Electrochim. Acta* 47 (2001) 577.
- [20] C.Y. Wang, W.B. Gu, B.Y. Liaw, *J. Electrochem. Soc.* 145 (1998) 3407.
- [21] V.R. Subramanian, V.D. Diwakar, D. Tapriyal, *J. Electrochem. Soc.* 152 (2005) A2002.
- [22] Y.-S. Lin, J. Reimers, *The 12th International Meeting on Lithium Batteries*, Nara, Japan, 2004.
- [23] J.I. Ramos, *Appl. Math. Comput.* 188 (2007) 739.
- [24] T.F. Fuller, M. Doyle, J. Newman, *J. Electrochem. Soc.* 141 (1994) 1.